

Detailed Report: Santana *Candidozyma auris* RNA-seq Reanalysis

Analysis/report cost: \$23.46

Executive summary

This report reanalyzes six paired-end bulk RNA-seq libraries from the Santana *Candidozyma auris* study. The primary comparison is AR0382_tnSWI1 versus AR0382; the optional comparison is AR0387 versus AR0382.

The central correction is identifier reconciliation by DIAMOND reciprocal protein matching. Paper/CGD SCF1 (B9J08_001458) maps exactly to current DE-table gene B9J08_03708, not to the superficially similar B9J08_01458. With this correction, SCF1 is strongly downregulated in both contrasts.

DIAMOND reciprocal protein matching

Paper/CGD protein	Current DE gene	Reciprocal?	Identity	Q cov	S cov	Bitscore
SCF1 / B9J08_001458	B9J08_03708	True	100%	100%	100%	1231
IFF4109 / B9J08_004109	B9J08_04863	True	100%	100%	100%	5076

Both key proteins are reciprocal best hits with 100% identity and 100% coverage in both directions. Unmasked DIAMOND was used for the final table because low-complexity adhesin sequence caused default masking to underreport IFF4109 coverage.

Corrected key-gene DE results

Contrast	Protein/gene	Current gene	Base mean	log2FC	padj	Interpretation
AR0382 tnSWI1 vs AR0382	SCF1 / B9J08_001458	B9J08_03708	23172.4	-6.817	0	Strong downregulation
AR0382 tnSWI1 vs AR0382	IFF4109 / B9J08_004109	B9J08_04863	898.9	-0.303	0.0392561505045197	Moderate downregulation
AR0387 vs AR0382	SCF1 / B9J08_001458	B9J08_03708	19427.8	-7.346	0	Strong downregulation
AR0387 vs AR0382	IFF4109 / B9J08_004109	B9J08_04863	667.4	-0.745	5.189722139454925400	Moderate downregulation

In the primary contrast, SCF1 normalized counts fall from approximately 46,989/44,890 in AR0382 to 384/427 in tnSWI1, corresponding to log2FC = -6.817. In the optional AR0387 contrast, SCF1 has log2FC = -7.346.

Figure 1. Primary contrast volcano plot

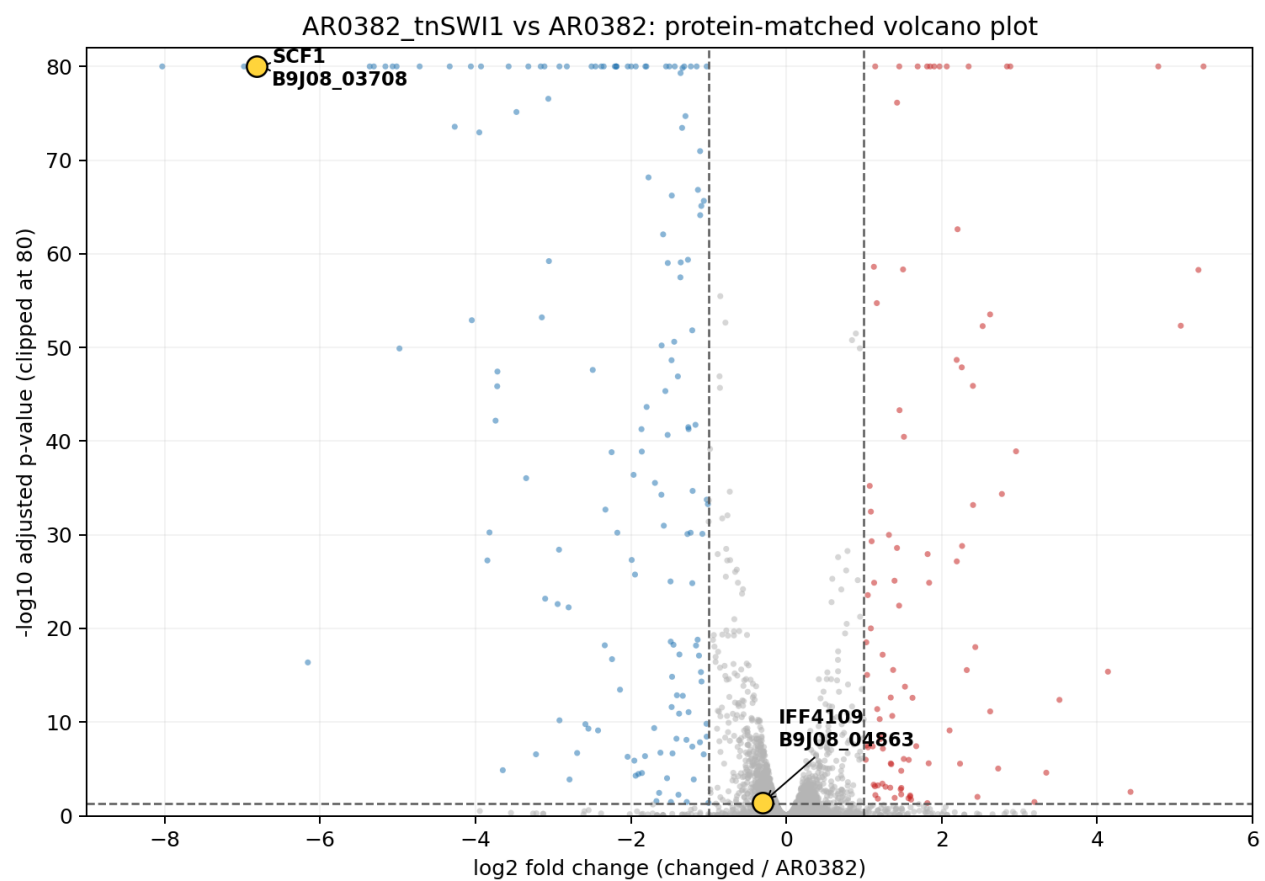


Figure 1. Volcano plot for AR0382_tnSWI1 vs AR0382. Yellow points highlight protein-matched SCF1 (B9J08_03708) and IFF4109 (B9J08_04863). SCF1 is among the strongest downregulated genes.

Figure 2. Optional AR0387 contrast volcano plot

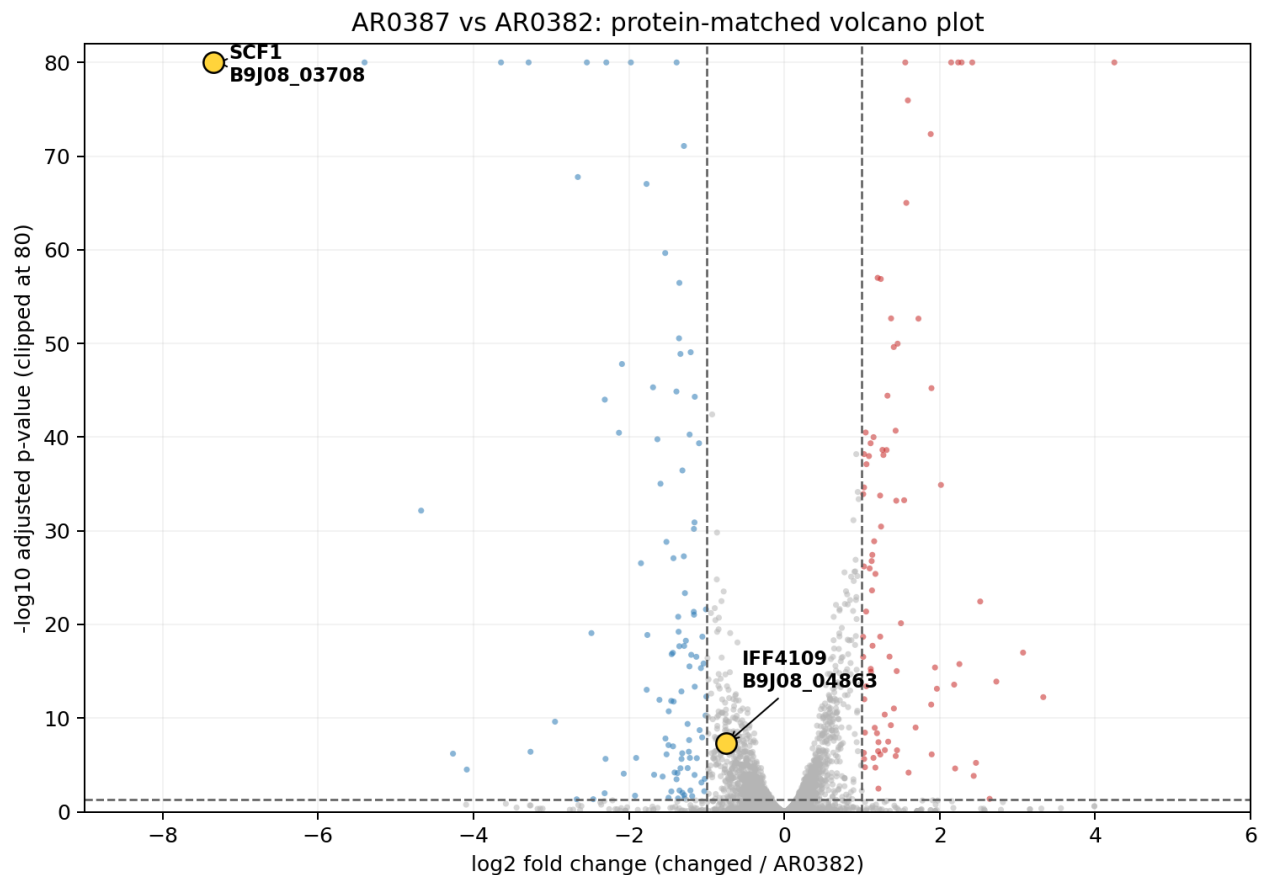


Figure 2. Volcano plot for AR0387 vs AR0382. Protein-matched SCF1 is again strongly downregulated, although this secondary contrast should be interpreted with more caution because AR0387_B had lower mapping metrics.

Interpretation

The corrected analysis supports the central biological interpretation from the Santana study: SCF1 is a major adhesin-associated transcriptional target whose expression is dramatically reduced in reduced-adhesion contexts.

The primary tnSWI1 result is the cleanest evidence. SWI1 disruption is associated with broad transcriptional remodeling and a particularly large collapse of SCF1 expression, consistent with SWI1-dependent regulation of an adhesin program.

The AR0387 comparison provides independent supporting evidence that low SCF1 expression is associated with a lower-adhesion isolate, but it should be framed as secondary due to lower mapping for AR0387_B.

IFF4109 remains relevant but is not the dominant signal in this reanalysis. Its protein-matched current gene B9J08_04863 is modestly downregulated in both contrasts, whereas SCF1 is dramatically downregulated.

Final conclusion

After DIAMOND reciprocal protein matching and image-supported DE review, this reanalysis reproduces strong SCF1 downregulation. The correct current gene ID for paper/CGD SCF1 is B9J08_03708. The corrected result aligns with the paper's conclusion that SCF1 is a major adhesin-associated factor linked to surface association, colonization, and virulence phenotypes.

Cost included as requested: \$23.46